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Q. Differentiate between ethics, law and professional responsiblity with suitable examples from the computing field.

Ans: Here are the differences between Ethics, Law and Professional Responsiblity:

Feature	Ethics	Law	Professional Responsiblity
Definition	Board moral principle that dictate what is 'right' vs 'wrong'.	Formal binding rules established by a government to maintain order.	Specific standards of conduct competence and duty expected of individuals.
Enforcement	Voluntary, guided by an individual's conscience, social norms or culture.	Mandatory; Enforced by government through legal penalties.	Enforced by professional org. or employers.
Computing Example	Choosing not to write a script that scrapes a competiting website	Complying with data privacy regulations or not hacking into bank's server.	Rigorously testing a medical software system before release to ensure there are no bugs to harm.

Q. What is Aristotle's Concept of the Golden Mean, and how does it guide individuals to find a balanced and virtuous course of action between extremes in ethical decision making?

Ans: The concept of the ~~g~~ Golden Mean was invented by Aristotle in his virtue ethics theory. According to him, virtue is the middle path between two ~~et~~ extremes:

- One extreme is excess
- The other extreme is deficiency

The golden Mean is the balanced point between these two extremes.

The Golden Mean helps individuals make balanced and thoughtful decisions.

① Avoiding Extreme Behavior: It teaches that extreme actions are usually harmful and unethical.

② Promoting Rational Thinking: Aristotle believed that people should use reason to determine the correct balance in each situation.

③ Encouraging Character Development: Instead of following strict rules, individual develop good character trails.

④ Application in Professional Life: Suppose a software engineer finds a mistake in a colleague's work:

- Deficiency: Ignoring the mistake completely.
- Excess: Publicly criticizing and embarassing him
- Golden Mean: Privately informing him and help to solve or fix the problem.

Q. Discuss the issue of moral decay in Bangladesh by analysing its major causes, social and economical affects and possible strategies to address and prevent it.

Ans: Moral decay refers to the declines in ethical values, honesty, discipline and responsibility within a society. It means people gradually ignore moral principle and act mainly for professional gain.

Major issues/causes of ~~Moray~~ Moral Decay in Bangladesh:

① Weak Moral and Religious Education: Lack of proper value-based education at home and school reduces awareness of honesty, respect and responsibility.

② Corruption and Lack of Accountability: Weak law enforcement and slow justice systems increase the problem.

③ Unemployment and Poverty: High unemployment and poverty sometimes push people toward unethical practices, such as bribery, fraud and crime.

Social decay of Moral Decay:

① Crimes such as theft, fraud - cybercrime may increase.

② People loss trust in institution, leader and event among themselves.

③ Young people may adopt harmful habit such as drug abuse.

Q. Explain the concept of green computing. How can computing professionals contribute to environmental sustainability?

Ans: Green computing refers to the environmentally reponsive design, use and disposal of computer and related systems. computers and data centers consume large amount of electricity. Electronic devices also produce hardful waste.

Environmental problems caused by it include:

- High energy usage
- Carbon emissions
- Toxic e-waste

Computing professionals play an important role in promoting environmental sustainbility.

① ~~Energy efficient~~

① Developers can write optimized code that consumes less power and processing.

② Organization can use energy-efficient devices and servers.

③ Using virtualization reduces the number of physical servers needed.

④ Old computers and electronic devices should be recycled properly.

⑤ Remote work and encouraging online meetings reduces travel and lowers carbon emissions.

Q. What is the conflict of interest in the IT workplace? Provide example and explain how professionals should ethically handle such situations.

Ans: A conflict of interest occurs when a person's personal interest interferes with their professional duties and responsibilities. In the IT workplace, this happens when an employee's private benefits, relationships or financial interest affects their ability to make fair and objective decisions.

Examples of conflict of interest in IT:

① Financial Interest: An IT manager select a software vendor owned by their relative even if another vendor offers better quality.

② Dual Employment: A software developer works for two competing companies at the same time and shares confidential knowledge.

③ Personal Gain: An employee uses company data or resources for personal business

④ Insider Information: A manager hires a friend or family member without proper qualification, ignoring more deserving candidates.

Ethical handling of conflict of interest:

① The important step is to openly inform the employer or management about any potential -conflict.

② If possible, the professional should remove themselves from decision-making in that situation.

③ Most organizations have ethical guidelines or codes of conduct. Professional must follow them.

Q. Discuss any two historical cases of ethical failure in the technology and explain the lessons learned from them.

Ans:

Case 1: Facebook - Cambridge Analytica Data Scandal:

In 2018, it was discovered that political consulting firm Cambridge Analytica collected personal data from millions of facebook users without their proper consent. The data was used to influence political campaigns and votes.

Ethical Issues: Misuse of personal data

Lack of tranceparecy

Failure to protect user information

Lesson Learned: Companies must protect user data carefully. Strong data protection law are in need. Ethical responsibility is as important as profit.

Case 2: Volkswagen Emissions Scandal:

In 2015, it was discovered that volkswagen installed special software in cars to cheat emission test.

Ethical Issues: Invetional software manipulation
Deception of regulators and customers
Environmental harm.

Lesson Learned: Technology must be not used to cheat. Companies must be ensure honesty. Strong internal ethical politics are necessary.

Q. Compare and Contrast Utilizationism and Deontology (Kantian ethics) as ethical frameworks.

Ans: Two important ethical theories are Utilizationism and B Deontology (Kantian ethics)

① Utilizorinism: An act is right if it produces the greatest happiness for the greatest number of people.

Key feature: Focuses on consequences (Result)
Decision depends on overall benefit

Example: If a software company shares user data to improve public safety and it ~~bef~~ benefits millions of people, a utiliarinism may say it acceptable.

② <u>Deontology (Kantian Ethics)</u>: An action is right if it follows moral rules, regardless of the consequences.

Key feature: Focuses on intention and moral duty
Some actions are always wrong.

<u>Example</u>: If sharing data benefits many people, it is wrong if it violates ~~re~~ privacy rights or breaks a promise.

Q. Explain virtue Ethics and describe how it can guide ethical decision-making for software engineers.

<u>Ans</u>: Virtue ethics fouses on a person's character rather than rules or consequences. It asks: "What kind of person should I be?" istead of "What rule should I follow?" According to this theory, a good

action comes from a good character. Important virtue in virtue Ethics; Honesty, Integrity, Responsibility, Fairness, Wisdom.

Virtue helps software engineers focus on building good character.

① A virtue engineer will not hide software bugs.

② An engineer who with strong moral character will protect user data and privacy.

③ A responsible engineer will test software properly before release, even if there is pressure.

④ A virtuous engineer will try to avoid bias and discrimination in AI system.

Q. What are the key principle of ACM code of ethics?

Ans: The a Association for Computing Machinery (ACM) code of Ethics provides guidelines for ethical behaviour in the computing profession. It helps computing professional act responsibly and protect society. Key principles are;

1. Contribute to society and human well-being.
2. Avoid actions that may cause physical, ph financial harm.
3. Be honest and trustworthy.
4. Be fair and take action not to discrimate
5. Respect privacy.
6. Sensitive information obtained during work must be kept secure.
7. Professional should continuously improve their skills.

Q. What is whistleblowing? Discuss the ethical considerations and challenges a computing professional might face ~~he~~ when deciding to blow the whistle.

Ans: Whistleblowing means reporting illegal, unethical or harmful activities within an organization to higher authorities, regulators or the public when deciding to blow the whistle.

A computing professional must think carefully about several issues:

1. If the ~~other~~ wrongdoing harms users, reporting it may protect the public.

2. Professional have a duty to be loyal to their organization.

(3) Many computing professionals handle ~~profess~~ confidential data.

(4) Before reporting, the professional must ensure the information is accurate.

(5) Whistleblowing should usually be consided after trying internal solutions.

Q: Explain the difference between copyright, patents and trademark in the context of software and digital products.

Ans:

Copyright: It protects the original expression of an idea. In software, it protects: source code, object code, website contents, graphics, audio, documentation.

Example: The source code of an OS like Microsoft Windows is protected by copyright. Others can't copy it.

Patents: It protects a new invention or technical solution. In software, patie patents may protect: Unique Algorithm, Technical processes, Innovative system method.

Trademarks: It protects brand names, logos and symbols that identify a product or company. In software and digital products, trademarks protects: Product name, company logos, Slogans.

Q. Compare open-source licensing with proprietary software licensing. What are the ethical implication of each model?

Ans: Software licensing defines how software can be used, modified and distributed. The two main models are:

① Open-source Licensing: It allows user to view, modify and distribute the source code freely under certain condition.

Example: The operating system Linux is open-source.

② Proprietary Software Licensing: This is owned by a company or individual. The source code is not shared with users.

Example: Microsoft windows and Adope Photoshop.

Ethical implications of open-source:

① Transparency; User can check the code for security.

② Collaboration; Encourages sharing knowledge and team.

Ethical implications of Proprietary Software;

① Intellectual property protection; Protects the developers right.

② Accountability; The company is clearly responsible for updates.

Q. Discuss the concept of informed consent in personal data collection. What ethical principle should organisations follow when collecting user data?

Ans: Concept of informed consent: It means that users clearly understand and agree to how their personal data will be collected, stored and shared.

In personal data collection, informed consent ensures that users know:

- What data is being collected
- Why it is being collected
- How long it will be stored
- Who will have access to it

Ethical principle organization should follow:

① Transparency: Organizations must clearly explain their data practices.

② Volume Consent: Consent must be freely given.

③ Purpose Limitation: Data should only be used for the purpose stated.

④ Data minimization: Only necessary data should be collected.

⑤ Privacy and Confidentiality: Organizations must protect personal data from the unauthorized access.

Q. What are the ethical concern associated with surveillance technologies? Explain with examples how these technologies can be misused.

Ans: Surveillance technologies are systems used to monitor people activities, behavior or communications. These includes:

- CCTV cameras
- Facial recognition system
- Internet tracking tools
- Phone monitoring system

Example of misuse:

① Political Monitoring: Surveillance tools can be used to monitor journalists, activists or political opponents

② Workplace Surveillance: Employee may excessively monitor employees' emails and activities, reducing trust and autonomy.

③ Social Media Tracking: Platforms may track user's online behavior to influence options or advertisement ~~with~~ without concern.

Q. Compare and contrast copyright and patents as forms of intellectual property protection in computing. What type of software asset is each best suited for and why?

Ans: Copyright: It protects the original expression of an idea, not the idea itself.

It protects: Source Code, Object Code, Website Design, Graphics, Audio.

Patents: It protects a new invention or technical process. It may protect: New Algorithms, Data processing methods, Hardware-Software integration techniques.

Patents are best suited for:

- Innovative technical methods
- Unique system design
- New technological solutions.

Because patents protect the underlying functional idea or invention not just written code.

Q. What is algorithmic bias? Explain its causes and suggest measure to ensure fairness in AI system.

Ans: Algorithmic bias refers to unfair discrimination in results produce by computer algorithms, especially in Artificial Intelligence system.

Cause of Algorithmic Bias:

① AI system learn from historical data. If the data contains social bias, the AI will learn and repeat.

② Incomplete or Unrepresentive data: If certain groups are underpesented in the dataset, the AI may perform poor.

③ Human Bias: Developers may unintentionally include their own assumptions while designing algorithm.

Measures to ensure Fairness in AI system:

① Use Training data should be different genders, races, ages.

② AI systems should be tested for discrimination before and after.

③ Develop systems that explain how decisions are made.

Q. Discuss the role of ethical leadership in IT organizations. How can leaders foster an ethical culture within their teams.

Ans: Ethical leadership means that leaders act with honesty, fairness, responsibility, and integrity. In IT organization, leaders play an important role because technology directly affects users, society and privacy.

Importance of Ethical Leadership:

① Leaders influence team behavior. If leaders act honestly and responsibly, they are more likely to follow the same values.

② IT leaders must be ensure that products are safe, secure and respectful of privacy.

~~③ Leaders can foster an Ethical Culture~~.

③ Accountability and Transparency

Leaders can Foster an Ethical Culture:

① Develop clear Ethical Policies

② Provide ethics training

③ Encourage open communication

Q. Discuss the importance of transparency and explainability in AI who should be held accountable when an autonomous system causes ~~heads~~ harm?

Ans: Importance of Transparency in AI:

① Build Trust: Users trust AI system when they understand how decisions are made.

② Prevents Misuse: Open information reduces hidden manipulation or unfair practices.

③ Project Rights: People can question decisions that affect.

Importance of Explainibility:

① Fairness: Helps detect bias

② Error Correction: If the system makes a mistake, it can be corrected.

③ Human oversight: Human can review and verify AI.

Accountability When an Autonomous System Cause Harm:

① **Developers and Engineers**: If harm is caused by poor design, coding errors or lack of testing.

② **Users and ~~Opet~~ Operators**: If the user misuses the system or ignores safety instructions.

Q. Explain the concept of responsible disclosure in cybersecurity. What ethical guidelines should a security researcher follow when discovering a vulnerability?

Ans: Responsible disclosure is the process of reporting a security vulnerability to the affected organization in a responsible and ethical way, before making the information public.

Responsible disclosure is important because;

- It prevents cybercriminals from exploiting the weakness.
- It gives the company time to release a patch or update.
- Protects user data and privacy.

Ethical guidlines for security researcher:

① Researcher should only test system where they have legal permission.

② ~~Report~~ The vulnerability should be reported directly to the company security system.

Q. Differentiate between ethical hacking and cybercrime. Under what conditions is hacking consided ethical?

Ans: Ethical hacking means legally testing a computer system to find security weakness and improve protection.

Purpose:

- To identify security vulnerability
- To prevent cyber attacks
- To protect data and system

Cybercrime means using computers or networks to commit illegal activities.

Purpose: • Stealing money or data

• Spreading malware

• Identity theft

Basis	Ethical Hacking	Cybercrime
Legality	Legal	Illegal
Permission	Done with permission	Done without permission
Purpose	Improve security	Personal gain or harm.
Intention	Protection	Exploitation

Hacking is considered ethical under the following condition :

① The hacker must have written permission from the system owner.

② The activity must follow Laws

③ The testing must stay within the agreed system boundaries.

Q. Why is explainability a key ethical requirement for AI systems used in high-stakes domains like credit lending or criminal justice? Discuss two social harms that can occur if ~~A~~ an AI system operates as a black box.

Ans: In high stack domains like credit lending or criminal justice, AI decisions profoundly impact human lives, freedom, and financial well-being. Explainability - the ability to understand ~~to~~ how an AI model arrived at a specific decision - is essential for accountability and due process.

Two Social Harms of "Black Box" AI Systems;

When an AI operates as a "Black Box", it can cause severe social harms:

Harm 1: A black box can secretly learn to base decisions on protected characteristics. (like race or gender) through proxy variables (like zip codes).

Harm 2: When an AI system makes a catastropic error (such as a false arrest based on facial recongnition), a black box prevents investigators from figuring out what went wrong. This creates a liability gap where no one can be held responsible.